
Rhythmic Timing Training

Low-Tech Protocol • Free Metronome App • No Special Equipment

A progressive, evidence-informed program for home, school, and clinic. Designed for pediatric occupational therapists, families, and educators.

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What This Is

This protocol is a structured, low-cost approach to rhythmic timing training using nothing more than a free metronome app and everyday materials. It works as a home program, a school-based strategy, a clinic activity progression, or a starting point before introducing more intensive technology-assisted approaches.

Free Metronome App Options

Recommended apps: **Metronome Beats** (free, Android/iOS) or **Pro Metronome** (free version). Any browser-based metronome works too — search "free online metronome" for no-download options.

Why Rhythmic Timing Training Works

The brain's internal timing system draws on the cerebellum, basal ganglia, and supplementary motor area working together. When this system is less efficient, it shows up as difficulty with bilateral coordination, motor planning, impulse control, processing speed, and reading fluency. Structured rhythmic training builds the precision of that timing system through repetition with immediate sensory feedback.

Area of Difficulty	How Timing Training Addresses It
Bilateral coordination	Trains left/right alternation and simultaneous limb control
Impulse control / ADHD	Requires holding back and waiting — the beat enforces inhibition
Motor planning & praxis	Builds sequenced, timed, multi-step movement programs
Processing speed	Increases speed of sensory detection and motor response
Reading & phonological processing	Temporal processing underlies phoneme discrimination
Attention & sustained focus	Demands continuous engagement throughout each session

Setup

- **Free metronome app** on any phone, tablet, or computer.
- **Everyday materials:** table surface, pencils or rhythm sticks, a ball (playground or tennis), floor tape for walking activities.
- **Positioning:** Seated with good postural support for upper-extremity tasks; standing or walking for whole-body activities.
- **Starting BPM:** Always begin slower than feels intuitive. 56–66 BPM is right for most children starting the program. Accuracy at a slow, steady tempo is harder than it looks and predicts performance at higher speeds.

How to Track Progress

At the end of a session, count how many beats the child tapped on time (within approximately half a second of the click). Divide by the total beats attempted, then multiply by 100 to get a percentage.

Progress Criteria

Formula: $\text{Accurate Beats} \div \text{Total Beats} \times 100 = \%$

Example: 17 accurate out of 20 beats = 85%

Progress rule: Advance to the next level when the child reaches 80% or higher across 3 of 4 sessions at the current level and BPM. Increase BPM by 4–8 at a time. Never rush the tempo progression.

**Level
1****Rhythm Awareness & Beat Detection***Listen before moving | 56–80 BPM***Beat Detection Clap-Along**

Play a metronome. Child listens for 8 beats silently, then joins in clapping every beat. Watch for rushing (ahead of the click) or dragging (behind it). Do not correct yet — just observe the pattern.

Timing

Attention

Metronome Lap Tap

Both hands tap together on the lap to a 60 BPM click. Goal: 20 consecutive taps within half a second of the beat. Progress to eyes closed once accurate with eyes open.

Timing

Attention

Processing Speed

Stomp the Beat

Standing, stomp one foot per beat. Progress to alternating feet once single-foot stomping is stable. Whole-body proprioceptive input anchors timing for many children.

Timing

Bilateral Coord

Proprioception

**Level
2****Bilateral Rhythm & Alternation***Left and right together | 60–72 BPM***Alternating Hand Taps**

Seated at a table. Right hand taps beat 1, left hand taps beat 2, alternating. Count accurate taps per 20-beat trial. Homolateral movement (same arm and leg on the same beat) is worth noting.

Timing

Bilateral Coord

Motor Sequencing

March to the Metronome

Each foot lift equals one beat. Arms swing opposite the stepping foot (contralateral pattern). Homolateral marching signals bilateral integration difficulty.

Timing

Bilateral Coord

Proprioception

Alternating Knee Pats with Midline Crossing

Right hand crosses to left knee, left hand crosses to right knee, on each beat. Adds interhemispheric demand to bilateral alternation.

Timing

Bilateral Coord

Midline Crossing

**Level
3****Motor Sequencing & Pattern Memory***Rhythm + sequence + recall | 60–80 BPM***Rhythm Stick Copycat**

Therapist taps a 4-beat pattern. Child waits, then repeats it exactly. Change the pattern each round. Progress from simple alternating patterns to syncopated rhythms with rests.

Timing

Motor
Sequencing

Praxis

Attention

Body Percussion Sequences

Build a 4-step sequence: clap hands, stomp foot, pat knees, snap. Child learns the sequence first, then performs it to a steady beat. Build to 6 and 8 steps as accuracy improves.

Timing

Motor
SequencingBilateral
Coord

Praxis

Call and Response

Therapist taps 2 beats within a steady 4-beat metronome framework. Child responds on beats 3–4 with a short pattern of their own. The metronome holds the shared timing reference throughout.

Timing

Motor
Sequencing

Attention

Processing
Speed

**Level
4****Anticipation & Internal Clock***Beat prediction without cue | 60–100 BPM***Silent Beat Continuation**

Child taps with the metronome for 8 beats. Metronome turns off. Child continues tapping for 8 beats alone. Metronome turns back on to compare. Score how many solo beats landed close to where the click would have been.

Timing

Internal Clock

Processing
Speed

Attention

Bounce the Ball on the Beat

Child bounces a playground ball to 60 BPM. The key is timing the release (anticipation), not just the contact. Progress to a tennis ball as control improves.

Timing

Internal Clock

Bilateral
Coord

Proprioception

Freeze and Predict

Child taps to a metronome. Pause it unexpectedly. Child counts silently and re-enters on the correct beat when the metronome resumes. Note how many beats off they were at re-entry.

Timing

Internal Clock

Attention

Processing
Speed

**Level
5****Dual-Task & Complex Timing***Timing while thinking | 70–120 BPM***Clap and Count**

Child claps to a beat while counting aloud by 2s, spelling a word, or naming days of the week. If rhythm falls apart when the verbal task begins, that is a dual-task deficit worth documenting.

Timing

Dual-Task

Attention

Processing
Speed**Heel-Toe Walking to a Beat**

Walk heel-to-toe along a floor tape line, one step per beat. Progress by adding: a verbal naming task, carrying an object, or catching a ball tossed on alternating beats.

Timing

Dual-Task

Bilateral
Coord

Proprioception

Name and Tap

Child alternates right/left taps on the beat while naming items from a category — animals, colors, foods — one name per beat. No materials needed. Scales to any difficulty by changing category or BPM.

Timing

Dual-Task

Processing
Speed

Attention

Session Log

Record the level, BPM, accurate beats, and total beats each session. Calculate percentage: $(\text{Accurate} \div \text{Total}) \times 100$.

Date	Level	BPM	Accurate	Total	%	Notes / Observations

Sample Goal Language

- Child will maintain rhythmic tapping within one beat of a 72 BPM metronome for 20 consecutive beats across 3 of 4 trials to support bilateral coordination and attention during academic tasks.
- Child will continue an internally generated 60 BPM beat for 8 beats without external auditory cue with no more than 1-beat error across 3 of 4 trials to support processing speed and motor planning.
- Child will complete a 4-step body percussion sequence to a 72 BPM metronome with fewer than 2 errors in 3 of 4 trials to address ideomotor praxis deficits impacting self-care and academic routines.

Rhythmic Timing & ADHD

ADHD involves differences in the basal ganglia — the same brain region central to internal timing. This is not coincidental. Research consistently shows that children with ADHD have measurable timing deficits independent of attention: they struggle to predict when a beat will arrive, they respond before the beat is fully processed, and their internal clock runs less consistently than neurotypical peers.

The Clinical Framing

Impulsivity is often a timing problem. "Rushing" on this protocol is the same mechanism as interrupting, acting before thinking, or starting a task before instructions are complete. The beat teaches the nervous system to wait, predict, and respond at the right moment — not just the first available moment.

How ADHD Changes Your Approach

Start extra slow

Children with ADHD tend to rush. Starting at 56 BPM makes the impulsive response feel obviously wrong, which accelerates self-awareness.

Shorter bursts

Keep trials to 10–15 beats instead of 20 when attention is fragile. Consistent short wins build more tolerance than long frustrating trials.

Eyes closed early

Removing visual input reduces distraction and forces the child to rely on the auditory beat. Counterintuitively, many ADHD kids do better eyes closed once they trust the beat.

Name the pattern

After a trial, ask: "Were you rushing or dragging?" Metacognitive labeling builds self-regulation. Use the same language every session.

Use Level 5 diagnostically

The "Clap and Count" exercise is particularly revealing with ADHD. If the clapping falls apart the moment a verbal task is added, that is dual-task processing data worth including in documentation.

What Progress Looks Like with ADHD

In-session accuracy scores are useful, but watch for these real-life indicators that timing training is transferring:

- Fewer interruptions during conversations or class discussions
- Improved transitions — less resistance to stopping and starting activities
- Better turn-taking in games and social situations
- Smoother alternating movements (buttoning, handwriting, cutting)
- Parent or teacher reports of "seeming more regulated" without a specific explanation

Parent FAQ

Answers to the questions families ask most often about this program.

Q: How long should each session be?

5 to 10 minutes daily is more effective than 30 minutes twice a week. The nervous system learns timing through repetition across many days, not long sessions. A short, consistent daily practice will outperform sporadic long sessions every time.

Q: My child hates doing this. Is that normal?

Yes. This is genuinely cognitively hard, especially for kids with timing difficulties. Expect resistance at the beginning. Start at a BPM that feels insultingly slow, keep sessions short, and celebrate percentage improvements rather than perfection. The frustration often decreases significantly by week 3 as the brain starts to adapt.

Q: How will I know it's actually working?

Look outside the sessions. The goal of timing training is not a high score on a clapping exercise — it is better bilateral coordination, smoother transitions, less impulsivity, and improved motor skills in daily life. Check in with your child's teacher, OT, or therapist every 4 weeks and ask whether they are noticing any of these changes.

Q: Do we have to do every exercise?

No. Work through the level in order when you first start, but once you know which exercises your child responds to, you can return to those. The level structure exists to build skills progressively — do not skip to Level 4 before Level 2 is solid.

Q: Do we need to do this every single day?

5 days per week is ideal. 3 days per week still produces results but slower. Skipping more than 2 consecutive days tends to reset some of the gains. Consistency matters more than duration.

Q: Can siblings do this together?

Yes, and it often goes better. The competitive element motivates some kids to stay on task longer, and taking turns gives each child time to observe the other's timing, which builds self-awareness. Just keep it collaborative rather than judgmental.

Q: When should I contact an OT?

If your child has been working consistently at Level 1 for 4+ weeks and has not reached 80%, if frustration is preventing engagement entirely, or if you are noticing significant difficulty with bilateral coordination, handwriting, or daily motor tasks — an individualized OT evaluation would give you much more specific guidance.

Clinical Disclaimer: This handout is for educational and informational purposes only. It does not constitute medical advice, diagnosis, or treatment. Rhythmic timing training is a supplemental approach and is not a substitute for individualized evaluation and therapy by a licensed occupational therapist or qualified provider. All sales are final. Technical issues: contact Admin@Evolve-OT.com within 48 hours.

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